

TURRET OS: Multi-Layered Graph and Rule-AST Architecture Rescuing Insider Threat Detection Against Out-of-Distribution Adversaries

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ABSTRACT

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1. Introduction
Insider incidents—abuse of authorised access by legitimate users—continue to be a leading operational, financial, and regulatory exposure for enterprise security programmes. The dominant defensive response today is machine-learning detection over user-behaviour features, file-system telemetry, and provenance graphs; however, when reduced to a single feature layer, these detectors degrade sharply under out-of-distribution adversarial adjustment, and their alert outputs rarely satisfy courtroom admissibility standards. This paper addresses two open problems in tandem: detector resilience under held-out out-of-distribution adversaries, and a continuous inference-to-evidence chain that delivers forensically admissible artefacts.

Prior research clusters in four strands. The first applies graph-neural architectures to multi-modal insider telemetry—SENTINEL [Xiao et al., 2024], DTGI [Gao et al., 2023], MEWRGNN [Xiao et al., 2022], and TGCN-DA [Li et al., 2023]—alongside feature-engineered baselines including Le and Zincir-Heywood [Le et al., 2020] and E-Watcher [Wei et al., 2024]. The second strand builds provenance-graph cybersecurity for tactical attack attribution—CSKG4APT [Ren et al., 2022], KRYSTAL [Kurniawan et al., 2022], KGRAG [Aki et al., 2025], and the provenance-graph survey of Li et al. [Li et al., 2021]—but typically terminates outputs at alert generation. The third strand extracts file-metadata signals: image-EXIF vectorisation in Exif2Vec [Umair et al., 2024] and Yasenenko [Yasenenko, 2025], and OOXML-revision and tamper forensics in Didriksen [Didriksen, 2014] and Fu et al. [Fu et al., 2015]. The fourth strand formalises digital forensic readiness under ISO/IEC 27043 for SaaS insider threats [Shoderu et al., 2025, 2026] and grounds provenance representation in W3C PROV [Yusof et al., 2025, Sikos and Philp, 2020] and operational trace frameworks [Garfinkel et al., 2013].

Several gaps remain. (i) Published detectors are predominantly single-layer—feature-engineered pipelines, monolithic graph neural networks, or stand-alone rule engines—and degrade sharply when faced with out-of-distribution adversaries whose observable features converge with benign

baseline traffic [Wei et al., 2024, Xiao et al., 2024, Gao et al., 2023]. (ii) File-metadata work remains narrowly restricted to image EXIF; cross-format provenance covering OOXML revisions, DWG authorship, EML headers, and git commit chains is absent [Umair et al., 2024, Yasenenko, 2025, Didriksen, 2014, Fu et al., 2015]. (iii) Provenance-graph research emits alerts but rarely couples them to W3C-PROV-aligned evidence chains suitable for forensic-readiness audits [Ren et al., 2022, Kurniawan et al., 2022, Aki et al., 2025, Li et al., 2021]. (iv) Readiness frameworks under ISO/IEC 27043 target SaaS-cloud deployments and have not been extended to on-premise or hybrid enterprise environments [Shoderu et al., 2025, 2026]. (v) No prior system braids multi-layer inference, cross-format metadata extraction, and court-grade evidence delivery in a single architecture.

Against these gaps we propose TURRET OS,¹ a four-layer architecture that fuses a cross-format metadata-harvest layer (L1), a Neo4j knowledge-graph layer (L2), a rule-AST inference layer (L3), and a multi-layer GraphSAGE inference layer (L3.5) into one calibrated decision surface; a fourth layer emits W3C-PROV-grounded, Merkle-chain-integrity-protected, Ed25519-signed evidence packs aligned to all eight ISO/IEC 27043 readiness attributes—Identification, Collection, Acquisition, Preservation, Analysis, Presentation, Chain of Custody, and Integrity Verification. Under an out-of-distribution credential-phishing adversary the multi-layer stack retains AUC 0.9572 with only a 4.12% drop, while single-layer feature detectors collapse to AUC 0.7363 with a 26.25% drop—a +22.09% architectural rescue. On the D5 Synthetic-Tenant Pilot (35 users, 90 days) with a held-out OOD profile, full-stack detection attains ROC-AUC 0.9928, F1 @ 0.5% FPR of 0.8936, and MCC 0.9539; the top SHAP feature is `copy_to_removable`, an operationally interpretable signal verified by a runtime feature-whitelist invariant that guarantees zero label leakage at deployment. End-to-end ISO/IEC 27043 readiness coverage is verified at 100% across all eight required attributes.

The remainder of the paper is organised as follows. Section 2 reviews prior literature across graph-neural insider detection, provenance-graph cybersecurity, file-metadata

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forensics, and ISO/IEC 27043 readiness. Section 3 specifies the adversary model, attack vectors, and evaluation criteria. Section 4 presents the TURRET OS architecture layer by layer. Section 5 reports the implementation stack on an 80-core, 503.58 GB-RAM, NVIDIA L40S, Python 3.12.9 platform. Section 6 details the experimental evaluation, including baseline comparison, layer ablation study, Pareto PR frontier, single-layer versus multi-layer adversarial robustness, ISO/IEC 27043 coverage, and compute budget. Section 7 discusses architectural and operational implications. Section 8 outlines limitations and future work. Section 9 concludes.

2. Literature Review

Earliest work on insider-threat detection framed the attacker as legitimate trusted credential misuse and pursued identification through hand-crafted rule heuristics and lightweight media forensics. [Garfinkel et al. \[2013\]](#) proposed media-based detection by content-aware fingerprinting and metadata-aware surface analysis of seized artefacts, demonstrating that visible media artefacts alone can discriminate threatening from benign insider behaviour. [Didriksen \[2014\]](#) examined the forensic artefacts of OOXML documents produced by Microsoft Office, showing that revision identity, custom-XML fingerprints, and document-version metadata can be assimilated into identification pipelines for document-level insider analysis. [Fu et al. \[2015\]](#) extended this line by analysing covert channels in Office 2007–2013 documents, demonstrating that XML structure and revision metadata can leak secrets to a co-leaked artefact. Limitations of this early media-only line were threefold: (i) coverage was restricted to a small set of document formats; (ii) operationally deployable detection was not demonstrated; and (iii) no automatic chain of forensic evidence was raised from the artefacts.

The next research wave widened the problem definition and laid the foundations of multi-modal detection and provenance representation. [Mavroeidis et al. \[2018\]](#) introduced a hybrid data-driven physical-security framework merging anomaly detection with user-behaviour profiling, demonstrating improvement of baseline detection on enterprise telemetry. [Le and Zincir-Heywood \[2019\]](#) and [Le et al. \[2020\]](#) systematically compared ensemble-ML and deep-learning approaches across multiple data-granularity levels of CERT insider datasets, with their longitudinal studies reporting machine-learning detection at AUROC 0.92 and F1 0.88 with time-to-detect of 14 minutes. [Homoliak et al. \[2018\]](#) catalogued the taxonomy of insider threats, attack vectors, and defensive controls in their seminal *Insight Into Insiders and IT*, establishing a reference map for the subsequent ML-detection lineage. [Sikos and Philp \[2020\]](#) surveyed provenance-aware knowledge representation models and contextualised knowledge graphs, advancing the formal foundations for representing and reasoning over cyber provenance traces. [Li et al. \[2021\]](#) consolidated system-level provenance-graph survey work for threat detection, establishing that the provenance-graph paradigm is the natural

substrate for post-hoc forensic reasoning over multi-modal telemetry. The limitation of this generation is structural: detection and provenance were treated as separate worlds, no system delivered both with court-grade evidence, and adversarial robustness was not yet measured through held-out attacker profiles.

The third research wave tied provenance-graph analytics to detection of advanced threats and assessed adversarial robustness experimentally. [Ren et al. \[2022\]](#) constructed CSKG4APT—a cybersecurity knowledge graph for advanced persistent threat organisation attribution—by fusing indicators of compromise, threat reports and malware artefacts into a unified graph. [Kurniawan et al. \[2022\]](#) introduced KRYSTAL—a knowledge-graph framework for tactical attack discovery in audit data—demonstrating graph-rule fusion for tactical-stage detection. [Xiao et al. \[2022\]](#) introduced MEWRGNN, a graph-neural-network architecture for anomaly-based insider-threat detection reporting AUROC 0.94 and F1 0.87. [Gao et al. \[2023\]](#) attacked the class-imbalance problem with Deep Temporal Graph Infomax and reported AUROC 0.91 and F1 0.86. [Li et al. \[2023\]](#) proposed TGCN-DA, a temporal graph convolutional network with data augmentation, and reported AUROC 0.95 and F1 0.89. [Xiao et al. \[2024\]](#) advanced the lineage with SENTINEL, a multi-timescale user-behaviour interaction graph learner reporting AUROC 0.93 and F1 0.88. [Wei et al. \[2024\]](#) reported E-Watcher with 98.48% accuracy and 99.23% F1 on CERT r4.2 on a 64 GB MacBook M1 Max platform, securing the most robust reported figures of the lineage but omitting adversarial robustness, cross-format provenance, and forensic-readiness attributes. [Mitra et al. \[2021\]](#) proposed provenance-based counter-measures against fake cyber-threat intelligence, demonstrating that provenance can guard the quality of processed alerts. The limitation of this generation is that detection capability and provenance representation deepened separately; adversarial robustness was not yet measured through held-out attacker profiles, no cross-format file-metadata vector was exploited, and operational evidence chains remained disconnected from ML alerts.

The most recent wave began treating metadata, provenance, and forensic-readiness as first-class research targets. [Umair et al. \[2024\]](#) proposed Exif2Vec, a framework to ascertain untrustworthy crowdsourced images by vectorising EXIF metadata, demonstrating that image-level provenance signals can expose tampering behaviour. [Yasenko \[2025\]](#) evaluated image-recognition systems for automatic detection of malicious files based on image metadata, reporting AUROC 0.76 on a curated image set. [Aki et al. \[2025\]](#) extended KGRAG—provenance-based cyber-threat-intelligence analysis with knowledge graphs and retrieval-augmented generation—pushing provenance representation into LLMs. [Yusof et al. \[2025\]](#) introduced PROVCON, an observability-to-provenance framework, demonstrating that operational telemetry can be elevated to W3C PROV provenance. [Shoderu et al. \[2025\]](#) and [Shoderu et al. \[2026\]](#) formalised digital forensic readiness under ISO/IEC 27043 for

the SaaS-cloud environment, but their evaluation was scoped to SaaS-cloud deployments and did not couple readiness to ML detection or on-premise/hybrid enterprise settings [Shoderu et al., 2025, 2026]. The remaining open gap of this generation is the absence of a unifying architecture that fuses cross-format metadata extraction, multi-layer joint inference, ISO/IEC 27043 readiness, and held-out attacker profiles within a single deployable system.

Synthesising the limitations of the four research waves—narrow format coverage of media forensics, separation of detection and provenance, absence of held-out adversarial regimes, and missing on-premise forensic-readiness coupling—we propose TURRET OS,² a four-layer architecture that fuses cross-format metadata harvest (L1), a Neo4j knowledge graph (L2), a rule-AST inference engine (L3), and multi-layer GraphSAGE inference (L3.5) into one calibrated decision surface; a fourth layer emits W3C-PROV-grounded, Merkle-chained, Ed25519-signed evidence packs aligned to all eight ISO/IEC 27043 readiness attributes. End-to-end detection with the multi-layer stack attains ROC-AUC 0.9997, F1 @ 0.5% FPR of 0.8774, and MCC 0.9770 on the D3 TSEC corpus; under an out-of-distribution credential-phishing adversary the multi-layer stack retains AUC 0.9572 with only 4.12% drop, while single-layer L1 feature detectors collapse to AUC 0.7363 with 26.25% drop—a +22.09% architectural rescue. On the D5 Synthetic-Tenant Pilot (D5, 35 users over 90 days, proactively tagged by Security Operations analysts) with a held-out OOD profile, full-stack detection attains ROC-AUC 0.9928, F1 @ 0.8936, and MCC 0.9539; SHAP feature-importance audit with a runtime whitelist invariant guarantees zero label leakage at deployment, with copy_to_removable as the dominant operational signal. ISO/IEC 27043 readiness coverage is verified at 100% across all eight attributes; computational budget is profiled with wall-clock timestamps on an 80-core, 503.58 GB-RAM, NVIDIA L40S, Python 3.12.9 deployment platform. TURRET OS thus closes the four limitations jointly, advancing the state-of-the-art from incremental single-layer detectors to a single deployable braided architecture with court-grade forensic evidence delivery.

References

- I. Aki, K. Watarai, S. Okada, et al. Provenance-based cti analysis with knowledge graph and rag. In *10th IEEE International Conference on Cyber Security and Cloud Computing (CSCloud)*, pages 8–15, 2025. doi: 10.1109/csccloud66326.2025.00008.
- E. Didriksen. Forensic analysis of ooxml documents. Master's thesis, Norwegian University of Science and Technology, 2014. URL <http://hdl.handle.net/11250/198656>.
- Z. Fu, X. Sun, and J. Xi. Digital forensics of microsoft office 2007–2013 documents to prevent covert communication. *Journal of Communications and Networks*, 17(1):91–99, 2015. doi: 10.1109/jcn.2015.000091.
- P. Gao, H. Zhang, M. Wang, et al. Deep temporal graph infomax for imbalanced insider threat detection. *INFORMS Journal on Computing*, 35(5):1020–1035, 2023. doi: 10.1080/08874417.2023.2267510.
- S. Garfinkel, N. Beebe, L. Liu, et al. Detecting threatening insiders with lightweight media forensics. In *IEEE International Conference on Technologies for Homeland Security (HST)*, pages 81–88, 2013. doi: 10.1109/ths.2013.6698981.
- I. Homoliak, F. Toffalini, J. Guarnizo, et al. Insight into insiders and it: A survey of insider threat taxonomy, detection, and defense. *arXiv preprint arXiv:1805.01612*, 2018. URL <http://arxiv.org/abs/1805.01612>.
- K. Kurniawan, A. Ekelhart, E. Kiesling, et al. Krystal: knowledge graph-based framework for tactical attack discovery in audit data. *Computers & Security*, 120:102828, 2022. doi: 10.1016/j.cose.2022.102828.
- D.C. Le and A.N. Zincir-Heywood. Machine learning based insider threat modelling and detection. In *IEEE International Conference on Communications (ICC)*, pages 1–6, 2019. URL <http://ieeexplore.ieee.org/document/8717892>.
- D.C. Le, A.N. Zincir-Heywood, and M.I. Heywood. Analyzing data granularity levels for insider threat detection using machine learning. *IEEE Transactions on Network and Service Management*, 17(1):412–425, 2020. doi: 10.1109/tnsm.2020.2967721.
- X. Li, L. Li, X. Li, et al. Tgcn-da: a temporal graph convolutional network with data augmentation for high accuracy insider threat detection. In *IEEE International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom)*, pages 170–178, 2023. doi: 10.1109/trustcom60117.2023.00170.
- Z. Li, Q.A. Chen, R. Yang, et al. Threat detection and investigation with system-level provenance graphs: a survey. *Computers & Security*, 106:102282, 2021. doi: 10.1016/j.cose.2021.102282.
- V. Mavroedis, K. Vishi, and A. Jøsang. A framework for data-driven physical security and insider threat detection. In *IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining (ASONAM)*, pages 850–857, 2018. doi: 10.1109/asonam.2018.8508599.
- S. Mitra, A. Piplai, S. Mittal, et al. Combating fake cyber threat intelligence using provenance in cybersecurity knowledge graphs. In *IEEE International Conference on Big Data*, pages 967–976, 2021. doi: 10.1109/bigdata52589.2021.9671867.
- Y. Ren, Y. Xiao, Y. Zhou, et al. Cskg4apt: a cybersecurity knowledge graph for advanced persistent threat organization attribution. *IEEE Transactions on Knowledge and Data Engineering*, 35(6):6123–6137, 2022. doi: 10.1109/tkde.2022.3175719.
- G. Shoderu, S.O. Baror, S. Makura, et al. Digital forensic readiness to mitigate insider threats in the saas cloud environment. *International Journal of Cyber Security*, 14(6):5059–5072, 2025. doi: 10.33022/ijcs.v14i6.5059.
- G. Shoderu, S. Baror, A. Modupe, et al. Digital forensic readiness to mitigate insider threats in the saas cloud environment. In *International Conference on Cyber Warfare and Security Proceedings*, volume 21, pages 4508–4516, 2026. doi: 10.34190/iccws.21.1.4508.
- L.F. Sikos and D. Philp. Provenance-aware knowledge representation: a survey of data models and contextualized knowledge graphs. *Journal of Data Semantics*, 9(3):118–135, 2020. doi: 10.1007/s41019-020-00118-0.
- M. Umair, A. Bouguettaya, A. Lakhdari, et al. Exif2vec: a framework to ascertain untrustworthy crowdsourced images using metadata. *ACM Transactions on Multimedia Computing, Communications, and Applications*, 20(4):3645094, 2024. doi: 10.1145/3645094.
- Z. Wei, U. Rauf, and F. Mohsen. E-watcher: insider threat monitoring and detection for enhanced security. *Annals of Telecommunications*, 79(5):1023–1038, 2024. doi: 10.1007/s12243-024-01023-7.
- F. Xiao, S. Chen, S.J. Chen, et al. Sentinel: insider threat detection based on multi-timescale user behavior interaction graph learning. *IEEE Transactions on Network Science and Engineering*, 11(4):3519155, 2024. doi: 10.1109/tnse.2024.3519155.
- J. Xiao, L. Yang, F. Zhong, et al. Robust anomaly-based insider threat detection using graph neural network. *IEEE Transactions on Network and Service Management*, 19(4):3222635, 2022. doi: 10.1109/tnsm.2022.3222635.
- V. Yassenenko. Evaluating the effectiveness of image recognition systems for automatic detection of malicious files based on image metadata. *Bulletin of Cherkasy State Technological University*, 2:77–85, 2025. doi: 10.62660/bcstu/2.2025.77.

²Project repository: <https://github.com/ali-Hamza817/Turret-OS>

- A. Yusof, S. Li, K. Singh, et al. From observations to insights: constructing effective cyberattack provenance with provcon. In *Workshop on Security Operations and Defense (WOSOC)*, page 23008, 2025. doi: 10.14722/wosoc.2025.23008.